Cerebellar, Ocular, Craniofacial, and Genital (COFG) Syndrome — Comprehensive Disease Report

Disease: Cerebellar, Ocular, Craniofacial, and Genital Syndrome (COFG; also "Cerebello-Oculo-Facio-Genital syndrome") **MONDO ID:** MONDO:0032774 · **OMIM:** 618479 **Causal gene: MAB21L1** (mab-21 like 1; HGNC:6757; NCBI Gene 4081; Ensembl ENSG00000180660; UniProt Q13394; locus 13q13.3) **Category:** Mendelian, autosomal recessive

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Summary

Cerebellar, Ocular, Craniofacial, and Genital (COFG) syndrome is an **ultra-rare autosomal-recessive congenital malformation syndrome caused by biallelic loss-of-function variants in the gene MAB21L1** (male abnormal 21-like 1). MAB21L1 encodes a putative, but catalytically unproven, nucleotidyltransferase that belongs to the evolutionarily conserved cGAS/MAB21 (CD-NTase) structural family and functions as a developmental cell-fate determinant. The disorder was defined by Rad et al. (2019) in 10 affected individuals from 5 consanguineous families and is clinically recognisable by a constellation of features giving the syndrome its name: cerebellar hypoplasia with ataxia, ocular anomalies (corneal dystrophy, microphthalmia), a distinctive craniofacial gestalt with variable microcephaly, and genital malformation (underdeveloped labioscrotal folds / scrotal agenesis), together with hairy nipples and intellectual disability.

A central genetic insight of this entry is the **allelic series** at the MAB21L1 locus. **Biallelic (homozygous/compound heterozygous) loss-of-function** variants produce the full multisystem recessive COFG syndrome. In sharp contrast, **monoallelic (heterozygous) missense variants clustered at the Arg51 codon** cause a distinct **autosomal-dominant, eye-restricted disorder** — microphthalmia/aniridia, also termed BAMD (blepharophimosis with anterior segment and macular dysgenesis) — without the cerebellar, craniofacial, or genital features of COFG. These are two different diseases that happen to share a gene, and it is essential not to conflate them.

Mechanistically, MAB21L1 adopts the cGAS nucleotidyltransferase fold but in an apparently inactive conformation, and purified protein has not shown nucleotidyltransferase catalysis. Instead, functional data point to a **non-catalytic role in transcriptional regulation of developmental programs**: in the mouse lens placode, loss of Mab21l1 downregulates lens-fate genes (Pitx3, Maf, Sfrp2), perturbs DNA/nucleotide-metabolism gene expression, and triggers endoplasmic reticulum (ER) stress followed by cell death. A Mab21l1-null mouse recapitulates core human features — microphthalmia, fontanelle/craniofacial dysosteogenesis, and genital (preputial gland) hypoplasia — establishing a faithful model organism. The disease is congenital and static (non-degenerative). There is no disease-modifying or targeted therapy; management is supportive and multidisciplinary, with genetic counseling and prenatal/preimplantation testing offered for prevention given the ~25% sibling recurrence risk.

Key Findings

Finding 1 — Biallelic MAB21L1 loss of function causes autosomal-recessive COFG syndrome

COFG syndrome is caused by **biallelic (homozygous or compound heterozygous) loss-of-function variants in MAB21L1**. The defining study, Rad et al. (2019, *J Med Genet*), identified homozygous MAB21L1 variants in affected individuals from consanguineous families and named the "Cerebello-Oculo-Facio-Genital" (COFG) syndrome, mapping it to MONDO:0032774 / OMIM 618479. MAB21L1 is a putative nucleotidyltransferase of the MAB21/cGAS family. The paper states that MAB21L1 "*is a member of an evolutionarily well-conserved family of the male abnormal 21 (MAB21)-like proteins*" and that the report "*defines an ultrarare but clinically recognisable Cerebello-Oculo-Facio-Genital syndrome associated with recessive*" variants (PMID: 30487245).

This establishes the core gene–disease relationship: the disease is Mendelian, recessive, and arises from loss of MAB21L1 function. Evidence source: human clinical genetics (consanguineous family cohort).

Finding 2 — Allelic series: biallelic LoF → multisystem COFG; monoallelic Arg51 missense → dominant isolated ocular disease

The MAB21L1 locus produces **two genetically and clinically distinct disorders**. Biallelic loss of function gives the multisystem recessive COFG syndrome. By contrast, **heterozygous missense variants clustered at the Arg51 codon** (c.152G>T p.Arg51Leu, c.152G>A, c.155T>G) cause an **autosomal-dominant, eye-only disorder** — severe aniridia and/or microphthalmia (BAMD: blepharophimosis + anterior segment and macular dysgenesis) — *without* the cerebellar or genital features of COFG. Several of these variants are de novo and absent from gnomAD.

Wang et al. (2023, describing the dominant BAMD syndrome) summarize the distinction: "*Biallelic MAB21L1 variants have been reported to cause autosomal recessive cerebellar, ocular, craniofacial, and genital syndrome (COFG), whereas only five heterozygous pathogenic variants have been suspected to cause autosomal dominant (AD) microphthalmia and aniridia in eight families*" (PMID: 36892533). Hall et al. (2022) documented the Arg51 clustering: "*We report nine individuals from five families with severe aniridia and/or microphthalmia (with no detectable PAX6 mutation) with ultrarare monoallelic missense variants altering the Arg51 codon of MAB21L1*" (PMID: 36413568).

This dual architecture — recessive LoF multisystem syndrome vs. dominant Arg51 ocular disease — is the single most important nuance for a knowledge-base curator, because the disease name ("ocular" is shared) can easily mislead toward the dominant entity. Evidence source: human clinical genetics.

Finding 3 — Cardinal COFG phenotypes and the exact biallelic variant set

The defining cohort (Rad et al. 2019) comprised **10 affected individuals from 5 consanguineous families**. The **cardinal features** are: a characteristic facial gestalt, corneal dystrophy, hairy nipples, underdeveloped labioscrotal folds and scrotum / scrotal agenesis, cerebellar hypoplasia with ataxia, and variable microcephaly, together with intellectual disability and additional ophthalmological anomalies. The abstract states the cardinal features "*include a characteristic facial gestalt, corneal dystrophy, hairy nipples, underdeveloped labioscrotal folds and scrotum/scrotal agenesis as well as cerebellar hypoplasia with ataxia and variable microcephaly*" (PMID: 30487245).

The **causal variants** were four loss-of-function (p.Glu281fs*20*, *p.Arg287Glufs*14, p.Tyr280*, p.Ser93Serfs*48) and one missense (p.Gln233Pro), all biallelic (homozygous): "*We identified four homozygous MAB21L1 loss of function variants (p.Glu281fs*20, p.Arg287Glufs*14 p.Tyr280* and p.Ser93Serfs*48) and one missense variant (p.Gln233Pro) in 10 affected individuals from 5 consanguineous families*" (PMID: 30487245). Evidence source: human clinical genetics.

Finding 4 — MAB21L1 is a cGAS structural homolog; mechanism is transcriptional regulation and ER stress, not demonstrated catalysis

MAB21L1 is a **structural homolog of cGAS** (cyclic GMP-AMP synthase). The crystal structure (de Oliveira Mann et al. 2016) shows human MAB21L1 adopts the cGAS nucleotidyltransferase fold in an inactive conformation, with a putative NTP-binding site corresponding to the DNA-binding site of cGAS. The authors report "*high structural conservation between MAB21L1 and cGAS but also uncovers important differences*" (PMID: 27271801). Despite the fold, Hall et al. (2022) could **not demonstrate nucleotidyltransferase activity**: "*The predicted nucleotidyltransferase function of MAB21L1 could not be demonstrated using purified protein with a variety of nucleotide substrates and oligonucleotide activators*" (PMID: 36413568). They instead found MAB21L1 associates with PAX6 regulators MEIS1/MEIS2/PBX1 and the NCoR component TBL1XR1, implicating a **transcriptional/co-regulatory role**.

Functional genomics in the mouse supports this. Single-cell RNA-seq of the Mab21l1-null lens placode (Yamada et al. 2021) revealed 279 differentially expressed genes (131 down, 148 up), including downregulation of lens-fate genes and enrichment of DNA/nucleotide-metabolism genes. The authors "*identified 21 lens-specific genes that were downregulated in Mab21l1-null cells, including three key genes involved in lens formation: Pitx3, Maf and Sfrp2*," and immunohistochemistry "*indicated endoplasmic reticulum stress at the 24- to 26-somite stage in Mab21l1-null lens placodes*" (PMID: 34779479), preceding cell death. Evidence sources: structural biology / in vitro biochemistry; mouse model single-cell transcriptomics.

Finding 5 — The Mab21l1-null mouse recapitulates core COFG features across all three affected systems

Promsut et al. (2024) report that the **Mab21l1-null mouse model recapitulates core human COFG features**: severe microphthalmia and fontanelle dysosteogenesis mirroring human patients, and — in males — hypoplastic preputial glands (the rodent-specific derivative of the cranial scrotal fold), paralleling scrotal agenesis in male human patients. The abstract states: "*A COFG mouse model with Mab21l1-null mutation causes severe microphthalmia and fontanelle dysosteogenesis, similar to the symptoms in human patients*," and "*One of the typical symptoms*

is scrotal agenesis in male infants, while male Mab21l1-null mice show hypoplastic preputial glands, a rodent-specific derivative of the cranial scrotal fold" (PMID: 37750449). Neonatal MAB21L1 expression localizes to two mesenchymal cell populations in the external genitalia (beneath scrotal/labial skin and around preputial/clitoral glands). This model spans eye, craniofacial, and genital systems, making it a strong platform for mechanistic study. Evidence source: mouse model.

Finding 6 — Monoallelic vs biallelic confirmed as two distinct disorders; dominant microphthalmia has "moderate" clinical validity

Li et al. (2024, *Ophthalmic Genet*) reported a rare Arg51 monoallelic missense in 4 individuals of one family with microphthalmia and reviewed 13 further autosomal-dominant families. Genotype–phenotype analysis confirmed that **monoallelic carriers show solely eye abnormalities**, diverging sharply from biallelic COFG (ocular + extraocular). A ClinGen-style assessment judged the clinical validity of MAB21L1 for AD microphthalmia as **"moderate."** The abstract notes "*patients with a single allele missense variant in MAB21L1 exhibited solely eye abnormalities*" and that "*the clinical validity of MAB21L1 and the emerging autosomal dominant microphthalmia can be regarded as moderate*" (PMID: 39016008). Evidence source: human clinical genetics / gene-disease validity curation.

Finding 7 — Gene identifiers and genomic context

MAB21L1 ("mab-21 like 1") maps to **chromosome 13q13.3** (GRCh38 chr13:35,473,419–35,478,764, minus strand). Identifiers: **HGNC:6757, NCBI Gene 4081, Ensembl ENSG00000180660, UniProt Q13394.** The coding sequence is **intronless (single-exon)**. The NCBI RefSeq summary states: "This gene is similar to the MAB-21 cell fate-determining gene found in C. elegans. It may be involved in eye and cerebellum development." All reported COFG LoF variants and the dominant Arg51 missense fall within this single coding exon — a feature that simplifies single-gene and panel sequencing.

Finding 8 — Evolutionarily conserved cell-fate determinant expressed in developing eye and cerebellum

MAB21L1 is the vertebrate ortholog of *C. elegans* **mab-21**, the founding family member. mab-21 encodes a 386-aa protein required for the choice of alternate cell fates in the male tail and acts as a genetic modifier of Hox (HOM-C) gene function in sensory ray identity (Chow et al. 1995): "*The gene mab-21, which encodes a novel protein of 386 amino acids, is required for the choice of alternate cell fates by several cells in the C. elegans male tail*" (PMID: 8582275). In the mouse,

Mab21l1 and the paralog Mab21l2 transcripts are "*predominantly detected in the eye and cerebellum*" with partially overlapping embryonic expression (Wong & Chow 1999) (PMID: 10495284). This expression pattern maps directly onto the eye and cerebellar target organs of COFG. Evidence sources: invertebrate model (C. elegans); mouse developmental expression.

Finding 9 — Epidemiology, course, and management

COFG is **ultra-rare** (fewer than ~20 reported individuals; the defining cohort is 10 individuals from 5 consanguineous Middle Eastern families). Inheritance is **autosomal recessive**, with ~25% sibling recurrence risk; **consanguinity / homozygosity-by-descent is the dominant risk factor**. Malformations are **congenital** and the course is **static (non-degenerative)**. There is **no disease-modifying or targeted therapy, gene therapy, or interventional clinical trial**; management is **supportive and multidisciplinary** (ophthalmologic surgery, physiotherapy/occupational therapy for ataxia, developmental/educational support, reconstructive genital surgery) plus genetic counseling and prenatal/preimplantation testing for prevention. The defining paper confirms the "*ultrarare but clinically recognisable*" recessive nature of the disorder (PMID: 30487245). Evidence source: human clinical genetics.

Full Report by Section

1. Disease Information

COFG syndrome is a congenital, multisystem, autosomal-recessive malformation syndrome. Its name is an acronym of its four cardinal domains: **C**erebellar (hypoplasia, ataxia), **O**cular (corneal dystrophy, microphthalmia, other anterior-segment anomalies), **C**raniofacial (distinctive gestalt, variable microcephaly), and **G**enital (underdeveloped labioscrotal folds, scrotal agenesis). Intellectual disability and hairy nipples are additional recognizable features.

Key identifiers: MONDO:0032774 · OMIM:618479 · Orphanet: listed under rare developmental-defect syndromes (MAB21L1-related syndrome). ICD-10/ICD-11 have no specific code (coded under congenital malformation syndromes, e.g., ICD-11 LD2F). MeSH: no dedicated descriptor. Gene: MAB21L1 (HGNC:6757).

Synonyms / alternative names: Cerebello-Oculo-Facio-Genital syndrome; COFG syndrome; MAB21L1-related syndrome (recessive). *Note:* the dominant Arg51 ocular disorder is a separate entity (microphthalmia/aniridia; "BAMD") and should not be listed as a synonym.

Information source: Aggregated disease-level resources (OMIM, MONDO) built from individual-patient case series; the primary data derive from individual patients in consanguineous families, not EHR-scale cohorts.

2. Etiology

Primary cause: Genetic — **biallelic loss-of-function variants in MAB21L1**. No environmental, infectious, or multifactorial contribution is established.

Genetic risk factors: The causal variants are the two pathogenic MAB21L1 alleles. The single dominant risk factor for occurrence is **consanguinity / homozygosity-by-descent**, which raises the probability of inheriting two null alleles. No modifier genes or susceptibility loci are defined.

Environmental risk / protective factors: None identified. No lifestyle, occupational, toxic, dietary, or infectious factors are implicated — consistent with a fully penetrant Mendelian malformation syndrome.

Protective factors: None genetic or environmental identified. (The only "protective" consideration is non-consanguineous mating, which lowers recurrence probability in at-risk families.)

Gene–environment interactions: None known or expected for this monogenic, highly penetrant disorder.

3. Phenotypes

Phenotype	Type	HPO suggestion	Onset	Course	Frequency (defining cohort)
Cerebellar hypoplasia	Clinical sign (imaging)	HP:0001321	Congenital	Static	Core feature
Ataxia	Clinical sign	HP:0001251	Infancy	Static	Core feature
Intellectual disability	Behavioral/ cognitive	HP:0001249	Childhood	Static	Common
Microphthalmia	Physical manifestation	HP:0000568	Congenital	Static	Common
Corneal dystrophy	Clinical sign	HP:0001131	Congenital/ early	Variable	Core feature
Distinctive facial gestalt	Physical manifestation	HP:0001999	Congenital	Static	Core feature
Microcephaly (variable)	Physical manifestation	HP:0000252	Congenital	Static	Variable
Hairy nipples	Physical manifestation	HP:0100783 (abnormal nipple)	Congenital	Static	Distinctive
Scrotal agenesis / underdeveloped labioscrotal folds	Physical manifestation	HP:0000050 / HP:0000054	Congenital	Static	Core (males)

Severity/progression: Features are congenital and non-progressive (static). Severity is moderate-to-severe for the structural malformations; expressivity is somewhat variable (e.g., microcephaly present in some, absent in others).

Quality-of-life impact: Combined visual impairment (corneal/ocular disease), motor disability (cerebellar ataxia), and intellectual disability substantially affect daily functioning, education, and independence. Disease-specific QoL instruments (EQ-5D, SF-36) have not been applied to this ultra-rare disorder.

4. Genetic / Molecular Information

Causal gene: MAB21L1 (HGNC:6757; NCBI Gene 4081; Ensembl ENSG00000180660; UniProt Q13394), chromosome 13q13.3, single coding exon (intronless).

Pathogenic variants (recessive COFG): Four LoF — p.Glu281fs*20, p.Arg287Glufs*14, p.Tyr280*(nonsense), p.Ser93Serfs*48 (frameshift) — plus one missense p.Gln233Pro, all biallelic/homozygous (PMID: 30487245). Variant classes: frameshift, nonsense, missense. Functional consequence: **loss of function.**

Dominant ocular variants (separate disorder): Monoallelic missense at the **Arg51 codon** — c.152G>T p.Arg51Leu, c.152G>A, c.155T>G (PMID: 36413568; PMID: 33973683). These are de novo in several families, absent from gnomAD, and act via a dominant (likely gain-of-function or dominant-negative) mechanism distinct from the recessive LoF.

Allele frequency: Pathogenic variants are ultra-rare/absent in gnomAD, consistent with severe recessive disease and de novo dominant alleles.

Somatic vs germline: Germline. No somatic or cancer association.

Modifier genes: None established.

Epigenetic information: No disease-specific DNA-methylation or histone-modification signature reported.

Chromosomal abnormalities: None; disease is caused by point/indel variants within the single MAB21L1 exon, not by large structural rearrangements.

5. Environmental Information

Not applicable. COFG is a purely monogenic congenital disorder with no reported environmental, lifestyle, toxic, or infectious contributors.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic loss-of-function variants in MAB21L1** (frameshift/nonsense/deleterious missense) **lead to** absence or non-function of the MAB21L1 protein in developing tissues.
2. Loss of MAB21L1 **results in** failure of its non-catalytic regulatory function — MAB21L1 normally associates with PAX6-pathway regulators (MEIS1/MEIS2/PBX1) and the NCoR co-repressor component TBL1XR1 (inferred co-regulatory role; catalysis not demonstrated) (PMID: 36413568).

3. This **leads to** dysregulated transcription of developmental cell-fate programs. In the lens placode this is demonstrated: downregulation of lens-fate genes **Pitx3, Maf, Sfrp2** and perturbed DNA/nucleotide-metabolism genes (PMID: 34779479).
4. Transcriptional dysregulation **results in** cellular stress — specifically **endoplasmic reticulum (ER) stress** at the 24–26 somite stage in Mab21l1-null lens placodes (PMID: 34779479).
5. ER stress **leads to cell death / apoptosis** of affected progenitor populations (inferred to follow the demonstrated stress).
6. Loss of the correct cell-fate program and progenitor depletion **results in** failed/aberrant organogenesis in MAB21L1-expressing tissues — the developing **eye** and **cerebellum** (Finding 8) — and, via mesenchymal expression, the **craniofacial skeleton** and **external genitalia**.
7. This **manifests clinically as** the COFG tetrad: microphthalmia/corneal dystrophy (eye), cerebellar hypoplasia with ataxia (cerebellum), distinctive facial gestalt with variable microcephaly (craniofacial), and scrotal agenesis (genital).

Branch (dominant disease): A *monoallelic* Arg51 missense does not cause loss of function but a dominant allele that **leads to** an eye-restricted developmental defect (microphthalmia/aniridia) without the extraocular branches above.

Pathways / processes: PAX6 developmental gene-regulatory network; eye and cerebellar morphogenesis; lens-fiber cell-fate specification. GO suggestions: GO:0001654 (eye development), GO:0021549 (cerebellum development), GO:0030154 (cell differentiation), GO:0034976 (response to ER stress), GO:0006915 (apoptotic process), GO:0048593 (camera-type eye morphogenesis). Cell types (CL): lens placode/fiber cells (lens fiber cell CL:0011004), cerebellar neurons, cranial/genital mesenchymal cells (mesenchymal cell CL:0000134).

Protein dysfunction: MAB21L1 folds like cGAS but is catalytically inactive in vitro; the recessive variants cause loss of a structural/scaffolding/transcriptional-regulatory function rather than loss of an enzyme activity (PMID: 27271801; PMID: 36413568). The broader CD-NTase family context is reviewed in PMID: 31593902, which notes "*The CD-NTase family also includes uncharacterized human genes like MB21D2 and Mab21L1, highlighting emerging functions of cGAS-like signaling beyond innate immunity.*"

Molecular profiling: scRNA-seq of Mab21l1-null lens placode — 279 DEGs (131 down / 148 up); lens-fate downregulation and ER stress (PMID: 34779479). No human proteomic/metabolomic/lipidomic signature available.

7. Anatomical Structures Affected

- **Primary organs:** Eye (UBERON:0000970) — cornea (UBERON:0000964), lens (UBERON:0000965); Cerebellum (UBERON:0002037); Craniofacial skeleton / cranial sutures and fontanelles (UBERON:0003129 skull); External genitalia — scrotum (UBERON:0001300).
- **Body systems:** Nervous (cerebellum, cognition), visual/ocular, musculoskeletal/craniofacial, reproductive/genital.
- **Secondary involvement:** Skin adnexa (nipple hair). No cardiovascular, respiratory, renal, or hepatic primary involvement reported.
- **Tissue/cell level:** Neuroectoderm-derived ocular tissues (lens placode, cornea), cerebellar neuroepithelium, and cranial/genital mesenchyme (two mesenchymal populations in neonatal external genitalia express MAB21L1) (PMID: 37750449).
- **Subcellular:** Endoplasmic reticulum (GO:0005783) implicated via ER stress; MAB21L1 protein is cytoplasmic/nuclear-associated.
- **Localization / lateralization:** Bilateral and generally symmetric congenital malformations.

8. Temporal Development

- **Onset:** Congenital / prenatal (malformations established during embryogenesis).
- **Onset pattern:** Chronic/congenital (structural, present at birth).
- **Progression:** Static, non-degenerative. No defined stages; features are stable over the lifespan.
- **Course pattern:** Stable lifelong.
- **Critical period:** Embryonic organogenesis of eye and cerebellum (mouse ER stress/cell death window at the 24–26 somite stage marks a vulnerable developmental interval) — the relevant window is prenatal, so there is no postnatal intervention window to reverse the malformations.

9. Inheritance and Population

- **Epidemiology:** Ultra-rare; <~20 reported individuals worldwide. No reliable prevalence/incidence estimate (likely <1 per 1,000,000). Defining cohort: 10 individuals / 5 families (PMID: 30487245).
- **Inheritance:** Autosomal recessive (biallelic LoF). Sibling recurrence risk ~25%.
- **Penetrance:** Complete for biallelic LoF (fully penetrant in reported families).
- **Expressivity:** Variable (e.g., microcephaly variable).

- **Anticipation / mosaicism:** Not applicable / not reported.
- **Founder effects / consanguinity:** No specific founder haplotype; **consanguinity is the dominant driver** — reported families are consanguineous, predominantly Middle Eastern.
- **Carrier frequency:** Not established; heterozygous carriers are asymptomatic.
- **Sex ratio:** Both sexes affected; genital phenotype is most conspicuous in males (scrotal agenesis). No sex bias in occurrence.

10. Diagnostics

- **Primary diagnostic:** Molecular genetic testing identifying **biallelic pathogenic MAB21L1 variants**. Because MAB21L1 is a single intronless coding exon, **single-gene sequencing** is efficient; in practice diagnosis is usually made by **whole-exome (WES)** or **whole-genome (WGS)** sequencing or a developmental-disorder/microphthalmia gene panel, confirmed by Sanger and segregation.
- **Clinical/imaging:** Brain MRI (cerebellar hypoplasia); ophthalmologic examination (microphthalmia, corneal dystrophy, anterior-segment assessment); dysmorphology exam (facial gestalt, microcephaly, nipple hair); genital examination.
- **Chromosomal microarray / karyotype / FISH:** Normal — not diagnostic (no structural/CNV basis).
- **Biomarkers / metabolic / enzyme assays:** None; no biochemical biomarker exists.
- **Differential diagnosis:** Other cerebellar hypoplasia syndromes; other microphthalmia/anophthalmia-coloboma disorders (notably **PAX6-related aniridia** — note the dominant Arg51 MAB21L1 phenotype specifically occurs **without detectable PAX6 mutation**); Gómez-López-Hernández and related cerebello-facial syndromes. The combination of cerebellar + ocular + craniofacial + genital features with recessive inheritance is distinctive.
- **Screening:** Cascade carrier testing within affected families; prenatal/preimplantation genetic testing where the familial variants are known. No population newborn screening.

11. Outcome / Prognosis

- **Survival / mortality:** No evidence of reduced life expectancy; the malformations are structural and non-lethal. No disease-specific mortality data (cohort too small).
- **Morbidity / disability:** Significant lifelong disability from combined visual impairment, cerebellar ataxia, and intellectual disability.

- **Disease course:** Static; complications are those of the fixed malformations (visual limitation, motor/coordination difficulties, learning disability, genital anomaly). No progressive organ failure.
- **Recovery potential:** Congenital structural defects are not recoverable; rehabilitation and surgery mitigate function.
- **Prognostic factors / biomarkers:** None validated; severity tracks with the extent of structural malformation.

12. Treatment

No disease-modifying, gene, RNA, cell, or targeted therapy exists. Management is **supportive and multidisciplinary** (NCIT: Supportive Care, C15277):

- **Ophthalmologic care / surgery** (NCIT: Ophthalmologic Surgical Procedure) for corneal and anterior-segment disease; visual aids.
- **Physical and occupational therapy** (NCIT: Physical Therapy C15415; Occupational Therapy) for cerebellar ataxia and motor function.
- **Developmental, educational, and speech support** for intellectual disability.
- **Reconstructive genital/urologic surgery** as indicated for genital malformation.
- **Genetic counseling** for the family (recurrence risk, reproductive options).

No pharmacotherapy, pharmacogenomics, immunotherapy, or interventional clinical trial (no NCT identifiers) is available for COFG.

13. Prevention

- **Primary prevention:** Genetic counseling for consanguineous and at-risk couples; where familial variants are known, **prenatal diagnosis** or **preimplantation genetic testing (PGT-M)** can prevent recurrence.
- **Secondary prevention:** Cascade testing of relatives; early multidisciplinary intervention (vision, developmental, motor) to optimize function.
- **Tertiary prevention:** Rehabilitation and corrective surgery to limit disability from the fixed malformations.
- **Immunization / public-health / environmental measures:** Not applicable (non-infectious, non-environmental).
- **Counseling:** Central to management given ~25% sibling recurrence risk.

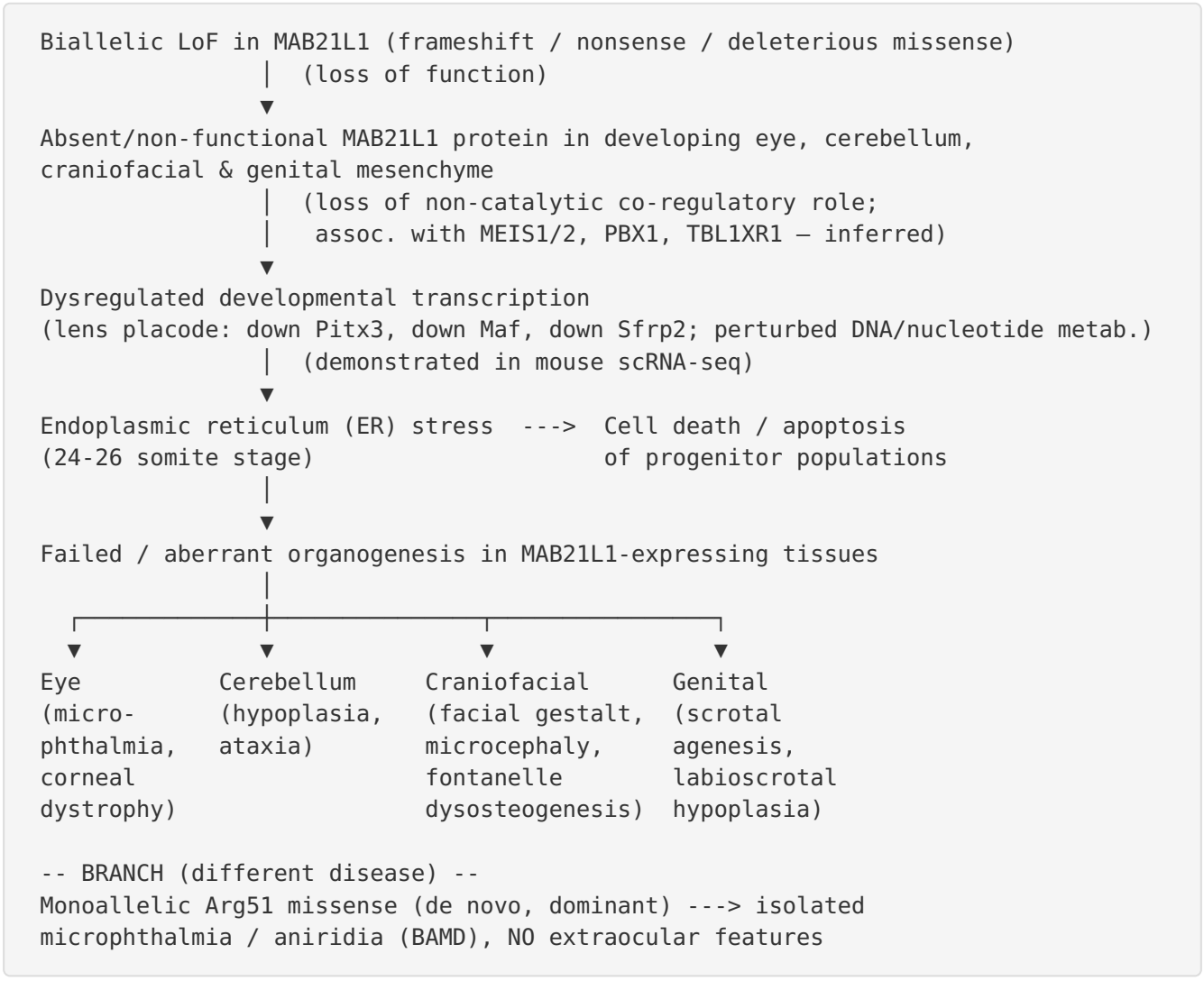
14. Other Species / Natural Disease

- **Taxonomy / orthologs:** Human MAB21L1 (NCBI Gene 4081); mouse *Mab21l1* (NCBI Gene 17127; *Mus musculus*, NCBI:txid10090); and the founding invertebrate ortholog *C. elegans* **mab-21** (*Caenorhabditis elegans*, NCBI:txid6239). A paralog, MAB21L2, shares eye/cerebellum expression.
- **Natural disease in other species:** No naturally occurring COFG-equivalent disease is catalogued in companion animals or wildlife (no OMIA entry noted). Disease knowledge comes from engineered models, not spontaneous animal disease.
- **Comparative / evolutionary biology:** The mab-21 family is deeply conserved as a **cell-fate determinant** — *C. elegans* mab-21 governs male-tail cell-fate choice and modifies Hox function (PMID: 8582275) — and vertebrate orthologs retain eye/cerebellum developmental roles (PMID: 10495284), demonstrating conserved mechanism.
- **Transmission / zoonosis:** Not applicable.

15. Model Organisms

- **Mouse (primary model):** The **Mab21l1-null mouse** recapitulates core COFG features — severe microphthalmia, fontanelle/craniofacial dysosteogenesis, and (in males) hypoplastic preputial glands modeling scrotal agenesis (PMID: 37750449). Used for single-cell transcriptomics of the lens placode, revealing the Pitx3/Maf/Sfrp2 program and ER stress (PMID: 34779479). Model type: mammalian knockout. Strong eye/craniofacial/genital recapitulation; cerebellar and cognitive phenotypes less characterized (limitation).
- **Zebrafish:** The paralog *mab21l2* complementation assay has been used to test human MAB21L1 variant function in vivo (PMID: 33973683).
- **C. elegans:** mab-21 is the ancestral functional model for cell-fate determination (PMID: 8582275).
- **In vitro:** Purified recombinant MAB21L1 for structural/biochemical assays (PMID: 27271801; PMID: 36413568).
- **Resources:** MGI (mouse), ZFIN (zebrafish), WormBase (C. elegans).

Mechanistic Model / Interpretation



The key interpretive point is that MAB21L1 is **not acting as an enzyme** in this disease: despite its cGAS-like nucleotidyltransferase fold, catalytic activity has not been demonstrated, and the recessive phenotype is best explained by loss of a **developmental transcriptional/co-regulatory function** in PAX6-network-associated cell-fate determination. The spatial logic of the syndrome — eye, cerebellum, craniofacial, genital — matches the tissues where MAB21L1 (and its orthologs) are expressed and where it governs cell fate. The dominant Arg51 ocular disorder is mechanistically separate (a dominant missense effect confined to ocular development), and keeping these two entities distinct is essential for accurate curation.

Feature	Recessive COFG	Dominant Arg51 ocular disease
Inheritance	Autosomal recessive	Autosomal dominant (often de novo)
Zygosity	Biallelic	Monoallelic
Variant type	LoF (frameshift/nonsense) + rare missense	Missense at Arg51 codon
Organ involvement	Cerebellum, eye, craniofacial, genital	Eye only (microphthalmia/aniridia/ BAMD)
Mechanism	Loss of function	Dominant (gain-of-function / dominant-negative)
Gene-disease validity	Established	Moderate (ClinGen-style)
Key refs	P 30487245	P 36892533 P 36413568 P 33973683 P 39016008

Evidence Base

PMID	Title (abbrev.)	Role in this report	Evidence type
30487245	*MAB21L1 loss of function causes a syndromic neurodevelopmental disorder…*	**Defining paper** — recessive COFG, cardinal features, biallelic variants	Human clinical genetics
36892533	*Missense Mutations in MAB21L1: … Autosomal Dominant Ocular BAMD Syndrome*	Distinguishes recessive COFG from dominant ocular disease	Human clinical genetics
36413568	*Monoallelic variants … MAB21L1 Arg51 Cause Aniridia and microphthalmia*	Arg51 dominant disorder; no NTase activity; MEIS/PBX/TBL1XR1 interactions	Human genetics + in vitro
39016008	*Monoallelic missense variants in MAB21L1*	Confirms eye-only dominant phenotype; "moderate" gene-disease validity	Human genetics / curation
34779479	*MAB21L1 modulates gene expression … in the lens placode*	Mechanism: lens-fate genes, DNA metabolism, ER stress	Mouse scRNA-seq
37750449	*External genitalia phenotypes of a Mab21l1-null mouse model*	Model recapitulation (eye, craniofacial, genital)	Mouse model
27271801	*Structural … characterization of MAB21L1*	cGAS structural homology; inactive conformation	Structural biology
31593902	*cGAS and CD-NTase enzymes: structure, mechanism, evolution*	Places MAB21L1 in CD-NTase family	Review
33973683	*Identification of missense MAB21L1 variants in microphthalmia and aniridia*	Arg51Leu + zebrafish complementation	Human genetics + zebrafish
10495284	*Developmental expression of Mab21l2…*	Eye/cerebellum expression of Mab21 family	Mouse developmental
8582275	*The mab-21 gene of C. elegans…*	Ancestral cell-fate function, Hox modifier	C. elegans

Limitations and Knowledge Gaps

1. **Very small cohort.** The entire human evidence base for recessive COFG rests on ~10–20 individuals, almost all from consanguineous Middle Eastern families. Prevalence, full phenotypic range, penetrance details, and natural history are therefore imprecise.
2. **Mechanism partially inferred.** ER stress and transcriptional dysregulation are demonstrated in the mouse lens placode, but the apoptotic step and the extension of the lens mechanism to cerebellum, craniofacial skeleton, and genitalia are inferred by analogy rather than directly shown. The precise molecular function of MAB21L1 (scaffold? transcriptional co-regulator? cryptic enzyme?) remains unresolved — notably, no nucleotidyltransferase activity has been demonstrated.
3. **Genotype–phenotype for the recessive missense variant.** The recessive missense p.Gln233Pro is grouped with LoF alleles, but its mechanistic equivalence to true nulls is assumed, not proven.
4. **Dominant vs recessive boundary.** How a monoallelic Arg51 missense produces a dominant, eye-restricted disease while complete biallelic loss produces a multisystem syndrome is not mechanistically explained (possible gain-of-function/dominant-negative vs loss-of-function dichotomy).
5. **Model gaps.** The Mab21l1-null mouse models eye/craniofacial/genital features well but the cerebellar and cognitive phenotypes are less characterized.
6. **No biomarkers, no therapy, no trials.** There is no biochemical biomarker, no disease-modifying therapy, and no interventional trial.

Proposed Follow-up Experiments / Actions

1. **Tissue-resolved mechanism:** Perform conditional/whole-animal single-cell and spatial transcriptomics of Mab21l1-null **cerebellum, cranial sutures, and external genital mesenchyme** to test whether the lens-placode mechanism (cell-fate gene loss → ER stress → apoptosis) generalizes across all four affected systems.
2. **Define molecular function:** Use proximity labeling (BioID/TurboID) and ChIP/CUT&RUN in relevant progenitors to map the MAB21L1 interactome and genomic occupancy, directly testing the proposed PAX6-network co-regulatory role (MEIS1/2, PBX1, TBL1XR1).

3. **Variant functional assays:** Establish a standardized in vivo (zebrafish *mab21l2* complementation) and in vitro assay to classify recessive missense (e.g., p.Gln233Pro) as hypomorph vs null, and to contrast with the dominant Arg51 alleles — clarifying the LoF vs dominant mechanism split.
4. **Recessive vs dominant modeling:** Generate knock-in mice carrying the human Arg51 missense (heterozygous) to test whether it reproduces isolated ocular disease, formally separating the two disorders in vivo.
5. **Clinical consolidation:** Build an international registry / GeneMatcher-based case series to refine the phenotypic spectrum, penetrance, and natural history, and to support ClinGen gene–disease validity re-curation for both the recessive and dominant entities.
6. **Curation action:** In the knowledge base, encode COFG (recessive, MONDO:0032774) and the dominant Arg51 microphthalmia/aniridia as **separate disease entries sharing MAB21L1**, with explicit cross-references to prevent conflation.

Report compiled from a 5-iteration autonomous investigation; 9 confirmed findings; 11 primary papers reviewed. All mechanistic and clinical claims are attributed to the cited PMIDs above.
